SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**

COURSE CODE: **CSA0501**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Technical Presentation (Low)

Assessment Tool Weightage: 35%

Code of Conduct:

I D. Gnana Deepthu(192411123)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS Total Marks: 50

Q.No	Question	Bloom's Level	Marks
1	Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.	BL2 – Understand	**5**
2	Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.	BL2 – Understand	**5**
3	Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.	BL2 – Understand	**5**
4	Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.	BL1 – Remember	**5**
5	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand	**5**
6	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand	**5**
7	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand	**5**
8	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember	**5**
9	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand	**5**
10	Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.	BL2 – Understand	**5**

RUBRICS FOR CALCULATION:

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals

Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0501 – DATABASE MANAGEMENT SYSTEM

TECHNICAL PRESENTATION

PRESENTED BY:

D. Gnana Deepthi (192411123)

1. DATABASE MANAGEMENT SYSTEM

- A Database Management System (DBMS) is software that creates, organizes, stores, retrieves, and manages data in a database.
- It acts as an intermediary between end users/applications and the physical database, hiding low-level storage details.
- Formally: DBMS = Database + set of programs to access and manage that data (define, construct, manipulate, share).
- Examples: Oracle, MySQL, PostgreSQL, Microsoft SQL Server, MongoDB.

OBJECTIVES OF A DBMS

- Data Abstraction & Independence — hide implementation details from users.
- Data Integrity — enforce rules/constraints so stored data remains accurate and consistent.
- Data Security — restrict unauthorized access through authentication and permissions.
- Concurrent Access — allow multiple users to access data simultaneously without conflict.
- Backup & Recovery — protect data against system or hardware failure.
- Minimizing Redundancy — avoid unnecessary duplication of data across files.

ADVANTAGES OF A DBMS

- Reduced Data Redundancy — centralized storage avoids duplicate copies of the same data.
- Data Consistency — updates are reflected uniformly across the system.
- Improved Data Security — role-based access control and authentication.
- Efficient Data Access — query languages (SQL) allow fast, flexible retrieval.
- Data Integrity — constraints (primary key, foreign key, checks) maintain correctness.
- Concurrent Access Control — transactions managed safely for many simultaneous users.
- Backup & Recovery Mechanisms — protects against data loss.

REAL-WORLD APPLICATIONS OF DBMS

- **Banking**

Account management, transactions, loan processing, fraud detection.

- **Healthcare**

Electronic Health Records (EHR), patient history, billing, lab systems.

- **Airlines & Railways**

Reservation systems, schedules, ticketing.

- **E-Commerce**

Product catalogs, orders, inventory, customer data.

- **Education**

Student records, admissions, examination results.

2. File-Based Systems vs DBMS

Aspect	File-Based System	DBMS
Data Redundancy	High — same data often duplicated across files	Minimized through centralization and normalization
Data Integrity	Difficult to enforce; scattered validation logic	Enforced centrally via constraints and rules
Data Security	Limited; file-level OS permissions only	Fine-grained, role-based access control
Concurrent Access	Prone to conflicts and data corruption	Managed via transactions and locking
Data Independence	Low — programs tightly coupled to file structure	High — logical and physical independence
Backup & Recovery	Manual, error-prone	Automated, built-in recovery mechanisms
Querying	Requires custom program for each query	Declarative query language (SQL)

CASE STUDY: BANKING — FILE-BASED VS DBMS

- **Legacy file-based branch banking (pre-1980s)**

- Each branch kept separate customer ledgers/files; no real-time link between branches.
- A customer could not deposit or check balance from a different branch ("branch banking", not "any-branch banking").
- Reconciliation between branches was manual and slow, increasing the risk of errors and fraud.

- **Modern DBMS-based Core Banking Systems (e.g. Finacle, Flexcube)**

- Centralized database lets any branch or ATM access the same live account record — enabling "anywhere banking".
- ACID transactions guarantee a fund transfer either completes fully or not at all, even during failures.
- Real-time fraud detection and audit trails are possible because all transactions sit in one consistent database.

Case Study: Healthcare — File-Based vs DBMS

- **Legacy paper/file-based patient records**

- Each hospital department (lab, pharmacy, billing) kept isolated paper or local files.
- A specialist in one department often could not see a patient's history from another department, risking duplicate tests or conflicting prescriptions.
- Records could be lost, damaged, or take a long time to retrieve in emergencies.

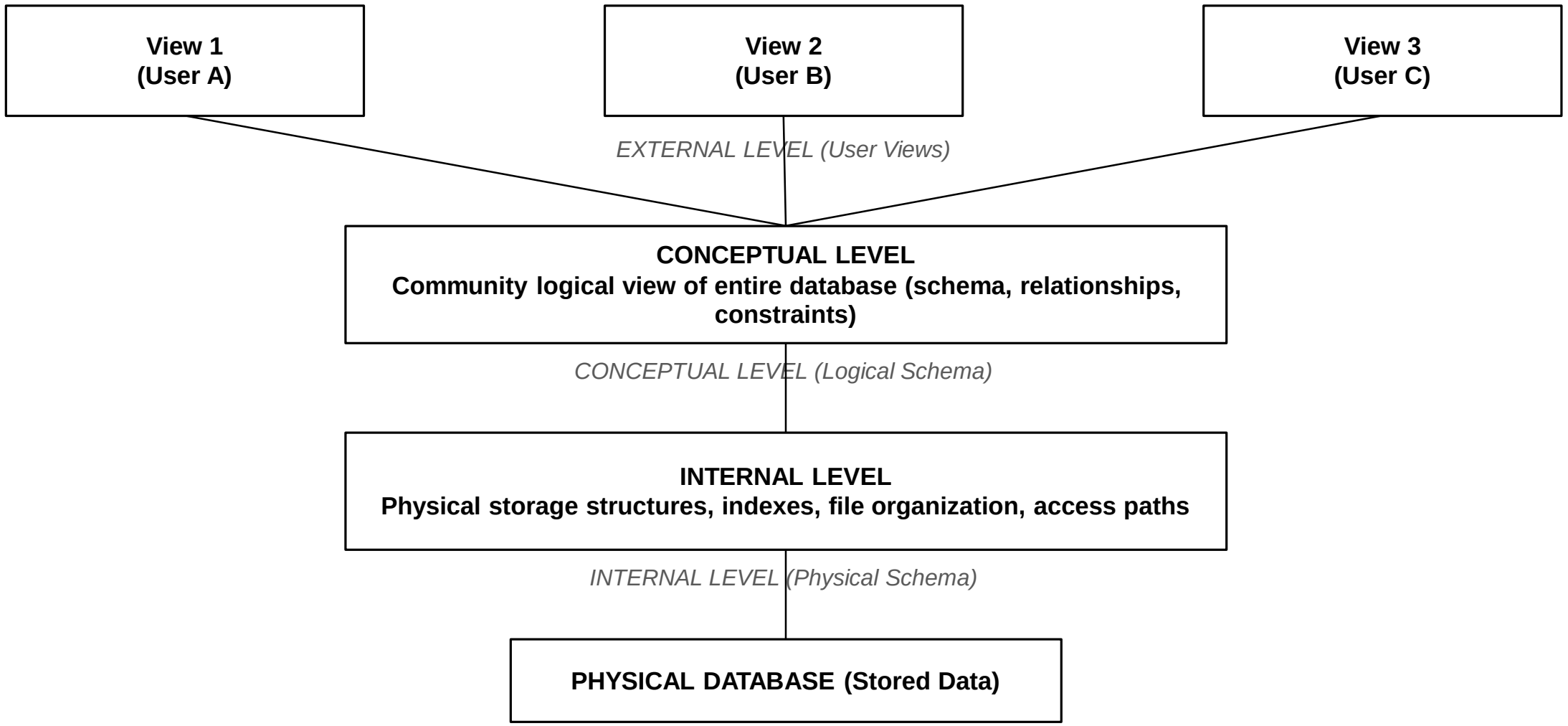
- **Modern DBMS-based Electronic Health Record (EHR) systems**

- A single patient record is shared securely across departments and even across hospitals in the same network.
- Access control ensures only authorized staff view sensitive data, satisfying regulations such as HIPAA.
- Instant retrieval of history, allergies, and medication improves emergency response and reduces medical errors.

3. THREE-LEVEL ANSI/SPARC ARCHITECTURE

- Proposed by ANSI/SPARC (1975) to separate user views from physical storage.
- Goal: provide data abstraction and data independence.
- Splits the system into three levels: External, Conceptual, and Internal.
- Each level describes the database at a different degree of abstraction.

Three-Level ANSI/SPARC Architecture — Diagram



Levels Explained with a Practical Example (University Database)

- **External Level**

- Registrar's view: student name, roll no., courses enrolled.
- Accounts office view: student name, fee status, scholarship details.
- Same underlying data, two different tailored views for two user groups.

- **Conceptual Level**

- One integrated schema: Student, Course, Enrollment, Fee tables with all relationships and constraints defined once.

- **Internal Level**

- How the Student table is actually stored: B+ tree index on roll number, file blocks on disk, storage in a specific format.

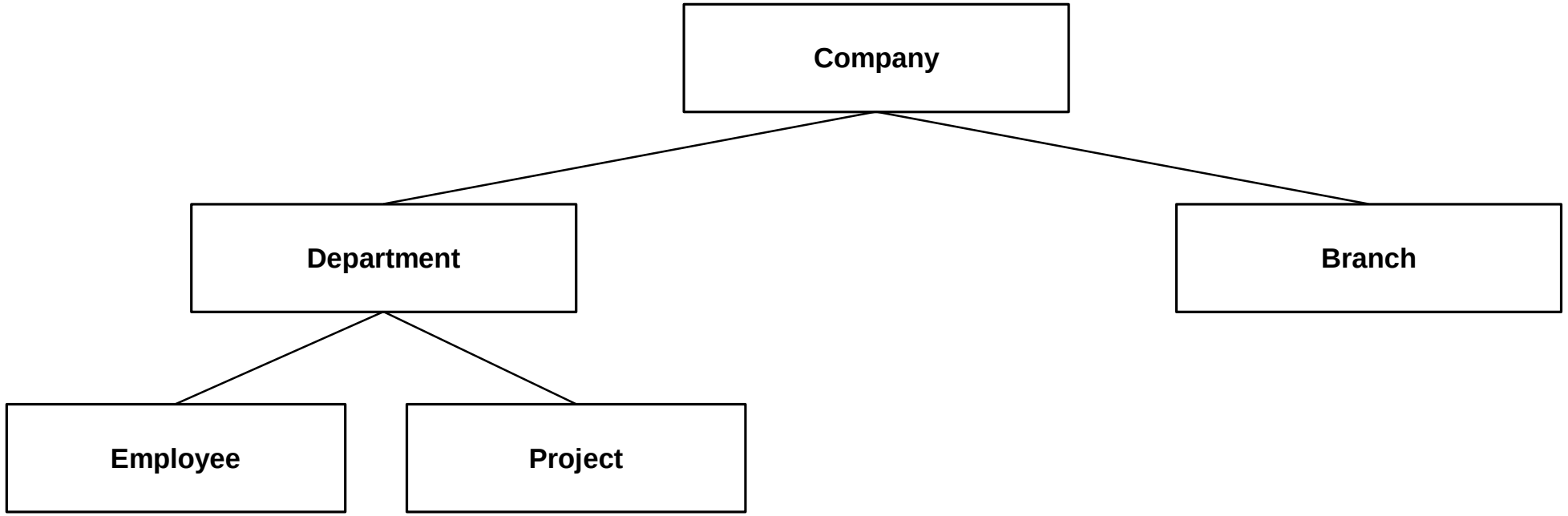
MAPPINGS BETWEEN LEVELS

- **External / Conceptual Mapping**
- Connects each external view to the conceptual schema; if the conceptual schema changes, only this mapping is updated — user views stay unaffected.
- **Conceptual / Internal Mapping**
- Connects the conceptual schema to the internal (physical) schema; allows physical storage to change without altering the logical structure.
- **These mappings are exactly what make logical and physical data independence possible.**

4. DATA MODEL

- A data model defines how data is logically structured, related, and manipulated within a DBMS.
- It provides the conceptual tools to describe data, data relationships, semantics, and constraints.
- Three classical models: Hierarchical, Network, and Relational (followed later by Object-Oriented and NoSQL models).

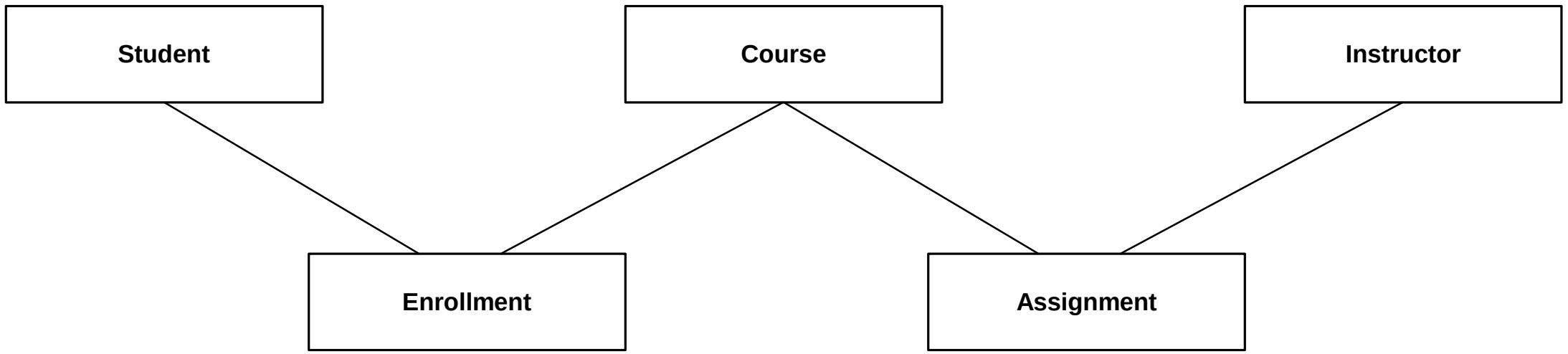
HIERARCHICAL DATA MODEL



Records are organized as a tree — each child has exactly one parent (one-to-many relationships only).

Industry Use Case: IBM's Information Management System (IMS) — still used in banking and airline mainframe systems for structured, high-volume transaction processing.

NETWORK DATA MODEL



Records can have multiple parents and multiple children — modeled as a graph, not strictly a tree (many-to-many relationships).

Industry Use Case: Integrated Data Store (IDS) and CODASYL-based systems historically used in early airline reservation and manufacturing inventory systems requiring complex interlinked records.

RELATIONAL DATA MODEL

Data is organized into two-dimensional tables (relations) of rows and columns; relationships are expressed using foreign keys — no navigational pointers needed.

Industry Use Case: Almost all modern business applications — banking (Oracle, DB2), e-commerce (MySQL, PostgreSQL), ERP systems (SAP HANA) — rely on the relational model for its simplicity and strong theoretical foundation (relational algebra, SQL).

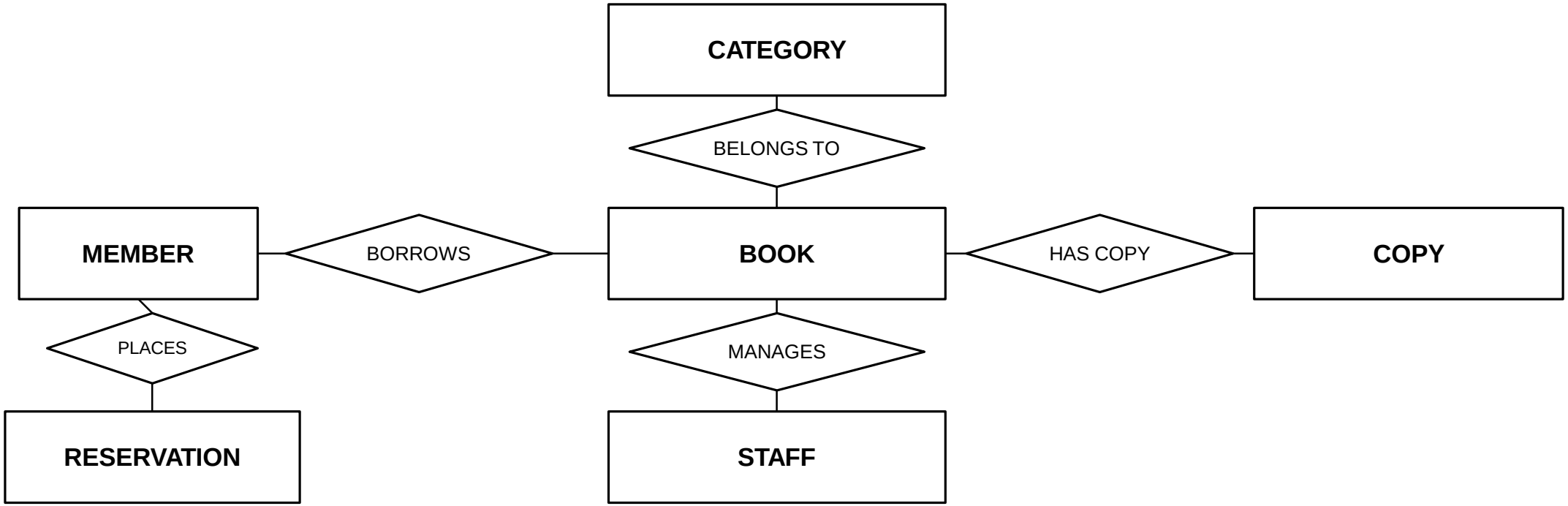
COMPARING THE THREE DATA MODELS

Aspect	Hierarchical	Network	Relational
Structure	Tree (parent-child)	Graph (owner-member)	Tables (rows & columns)
Relationships	One-to-many only	Many-to-many supported	Any, via foreign keys
Navigation	Top-down traversal	Pointer-based navigation	Declarative (SQL) queries
Flexibility	Rigid, hard to restructure	Complex but flexible links	High flexibility, easy to alter schema
Example System	IBM IMS	IDMS / CODASYL	Oracle, MySQL, PostgreSQL

5. ONLINE LIBRARY SYSTEM

- Members can browse, search, and borrow Books.
- Each Book belongs to a Category and may have multiple Copies.
- A Member can issue multiple Books over time; each issue is recorded as a Loan transaction.
- The library employs Staff who manage the catalog and process returns/fines.
- The system tracks Reservations for books currently on loan.

ER DIAGRAM — ONLINE LIBRARY SYSTEM

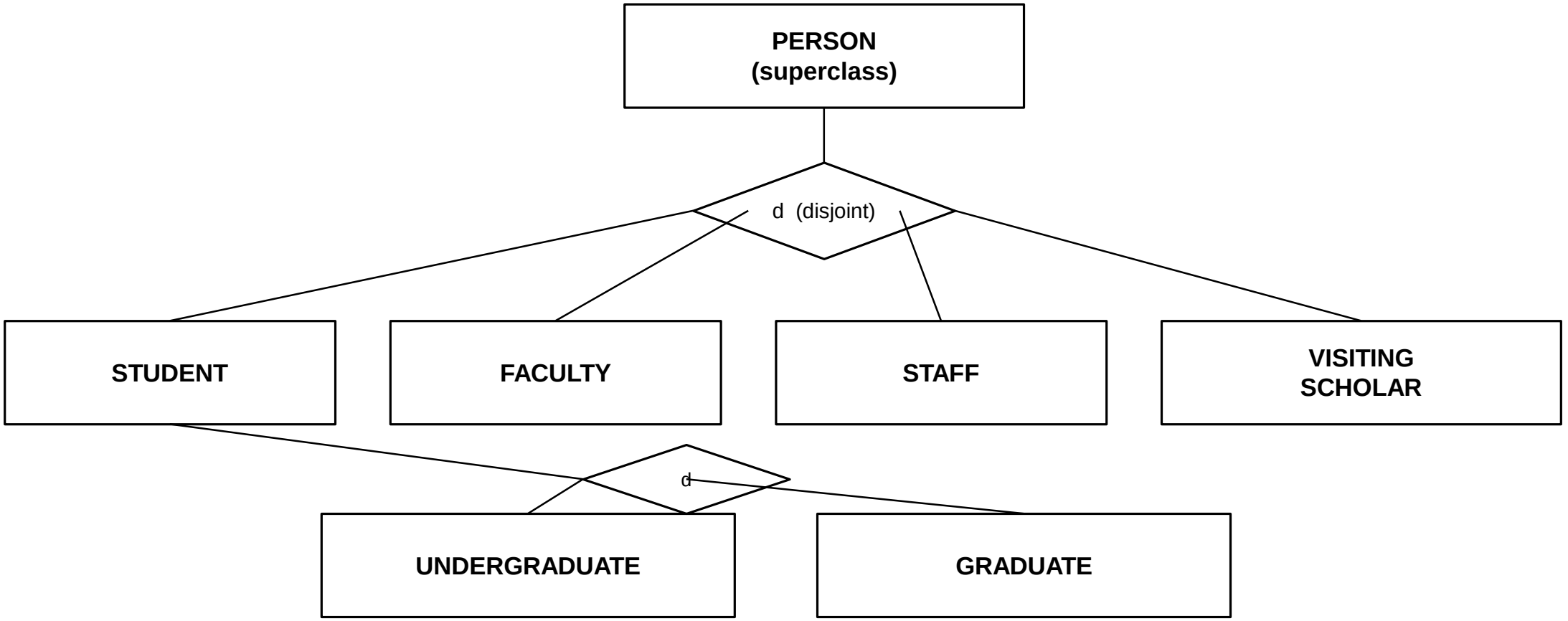


Rectangle = entity · Diamond = relationship. A Reservation records which Book a Member has requested.

6. THE ENHANCED ER (EER) MODEL

- Extends the basic ER model with concepts needed for more complex, real-world databases.
- Adds: Generalization, Specialization, Categorization (Union), and Attribute/Relationship inheritance.
- Generalization — bottom-up process: combine lower-level entities that share common attributes into a higher-level (super-class) entity.
- Specialization — top-down process: divide a higher-level entity into lower-level (sub-class) entities based on distinguishing characteristics.

University Case Study — Generalization / Specialization



PERSON specializes into STUDENT / FACULTY / STAFF / VISITING SCHOLAR (disjoint, based on "Role"). STUDENT further specializes into UNDERGRADUATE / GRADUATE.

7. LOGICAL DATA INDEPENDENCE

- The capacity to change the conceptual (logical) schema without requiring changes to external schemas or application programs.
- Example: Adding a new attribute (e.g. Email) to the Student table, or adding a new entity, does not force existing application queries to be rewritten.
- Achieved through the External / Conceptual mapping in the ANSI/SPARC architecture.
- Generally harder to achieve than physical independence, since logical changes can affect the meaning of data seen by users.

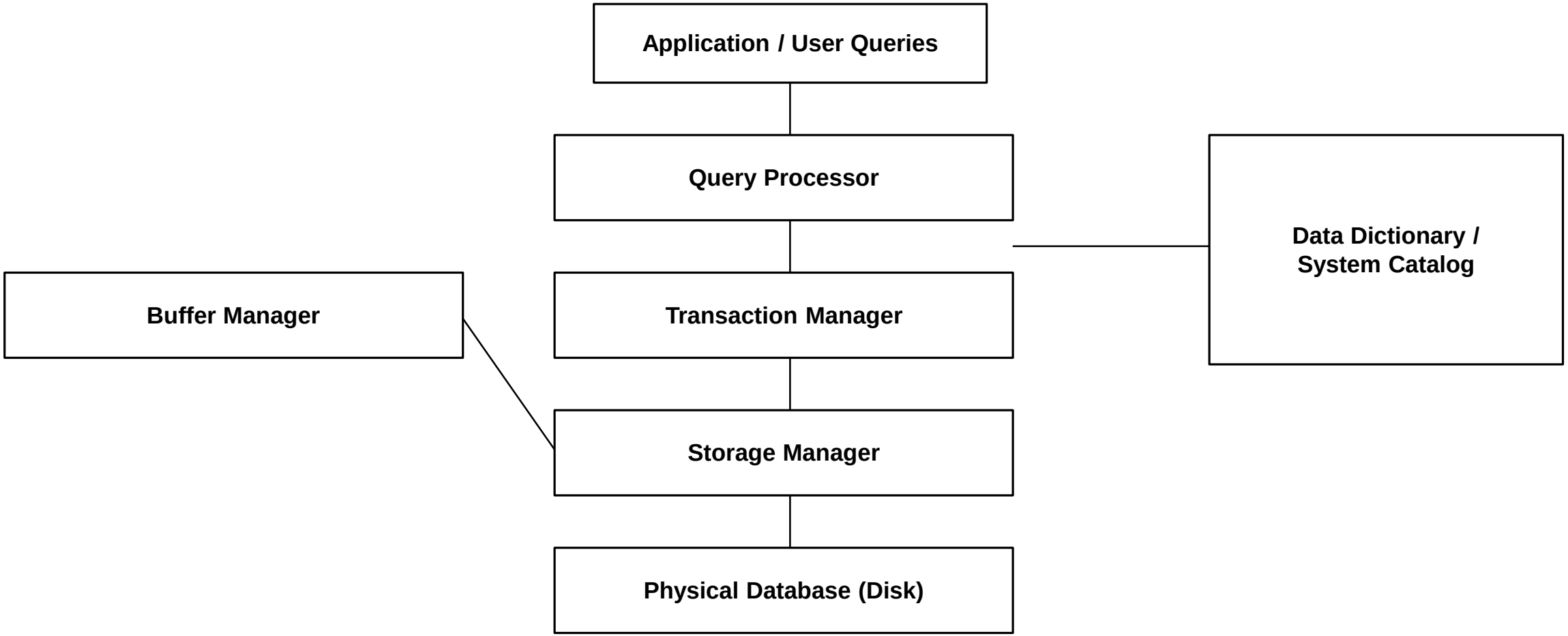
PHYSICAL DATA INDEPENDENCE

- The capacity to change the internal (physical) schema — storage structures, file organization, indexing — without altering the conceptual schema or application programs.
- Example: Switching a table's storage from a heap file to a B+ tree indexed file, or moving data to new disk hardware, is invisible to applications.
- Achieved through the Conceptual / Internal mapping in the ANSI/SPARC architecture.
- Easier to achieve than logical independence, because it does not affect the logical meaning of the data — only how it is stored.

SIGNIFICANCE OF DATA INDEPENDENCE IN DBMS ARCHITECTURE

- Reduces maintenance cost — schema or storage changes do not ripple through every application.
- Supports evolution — databases can grow and be optimized over time without breaking existing software.
- Enables true separation of concerns across the three ANSI/SPARC levels (external, conceptual, internal).
- Foundational goal that distinguishes a modern DBMS from simple file-processing systems.

DBMS Software Architecture — Component Overview



QUERY PROCESSOR

- Translates high-level queries (SQL) into low-level instructions the storage manager can execute.
- **Key sub-components:**

DDL Interpreter — processes schema definitions (CREATE, ALTER).

DML Compiler — translates DML statements into an execution plan (relational algebra expression).

Query Optimizer — chooses the most efficient execution strategy among alternatives.

Query Evaluation Engine — executes the chosen plan and returns results.

STORAGE MANAGER

- Bridges the gap between the application/query layer and the physical operating-system files.
- **Key sub-components:**

Authorization & Integrity Manager — checks access permissions and integrity constraints.

File Manager — manages allocation of disk space and the file structures used to store data.

Buffer Manager — decides which data to cache in main memory for fast access.

Ensures data is read from and written to disk efficiently and correctly.

TRANSACTION MANAGER

- Ensures the database remains in a consistent state even with concurrent access and system failures.
- **Responsible for the ACID properties:**

Atomicity — a transaction executes completely or not at all.

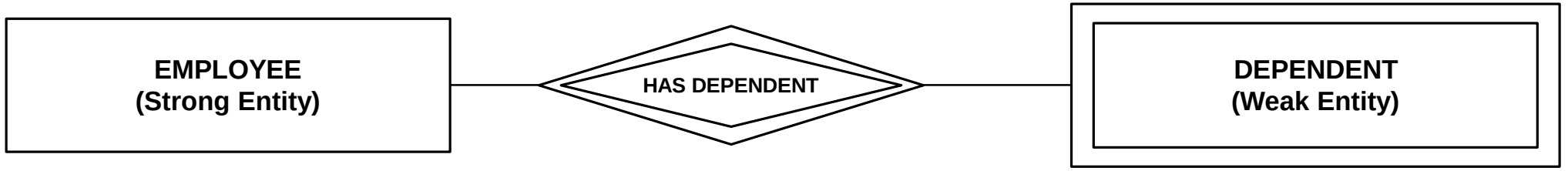
Consistency — a transaction moves the database from one valid state to another.

Isolation — concurrent transactions do not interfere with each other's intermediate results.

Durability — once committed, changes persist even after a crash.

Includes the Concurrency-Control Manager and the Recovery Manager.

9. STRONG ENTITY VS WEAK ENTITY — ER NOTATION



Single-line rectangle · has its own primary key

Double-line rectangle · no primary key of its own

The double diamond is an Identifying Relationship — EMPLOYEE (owner) is required for DEPENDENT to exist.

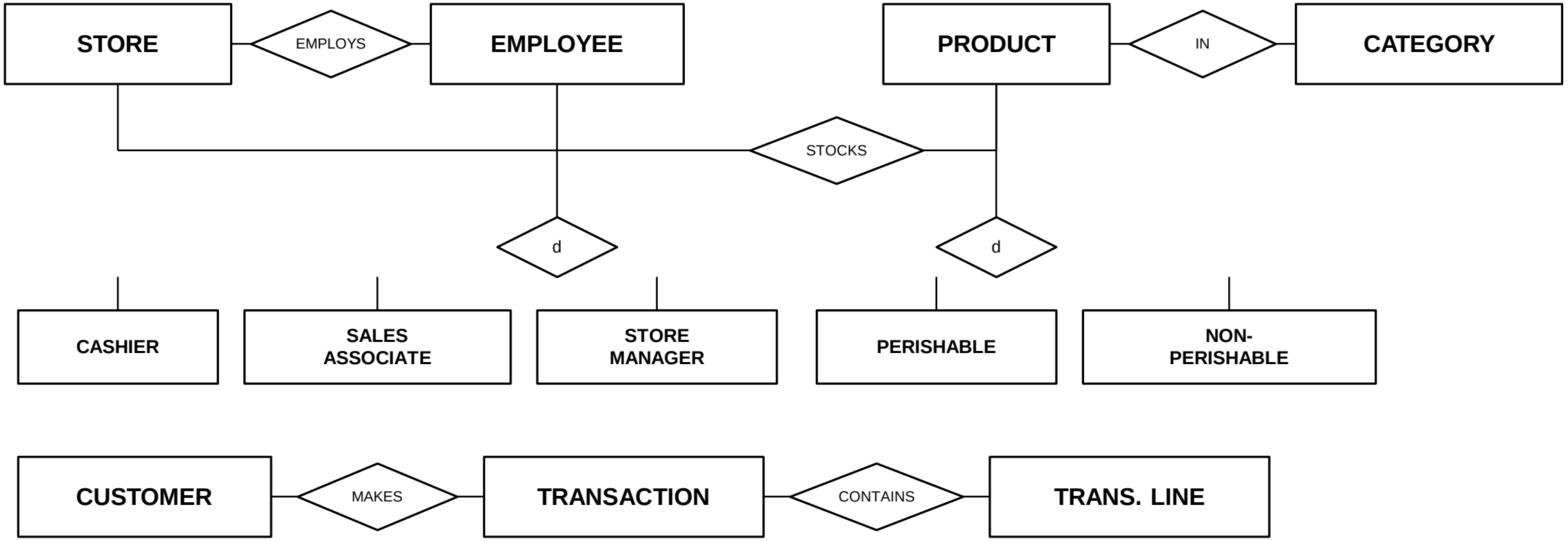
Strong vs Weak Entities — Comparison & Real-World Examples

Aspect	Strong Entity	Weak Entity
Primary Key	Has its own unique primary key	No primary key of its own — uses a partial key + owner's key
Existence	Exists independently	Existence depends on a related (owner) strong entity
ER Notation	Single-line rectangle, single diamond	Double-line rectangle, double diamond (identifying relationship)
Example 1	EMPLOYEE (Emp_ID as PK)	DEPENDENT (depends on Employee, uses Dependent_Name + Emp_ID)
Example 2	ORDER (Order_ID as PK) in e-commerce	ORDER_ITEM (identified only via Order_ID + line number)
Example 3	HOTEL (Hotel_ID as PK)	ROOM (Room_No is unique only within a specific hotel)

10. Retail Chain Management System — Requirements

- A retail company operates multiple Stores across different cities, each with its own Manager.
- Each Store sells Products organized under Product Categories, and maintains its own Inventory.
- Customers make Purchases; each purchase is a Transaction containing one or more Products.
- Employees are specialized into Cashiers, Sales Associates, and Store Managers, each with distinct attributes.
- Products are specialized into Perishable Goods (with an expiry date) and Non-Perishable Goods (with a shelf life category).

EER Diagram — Retail Chain Management System



d = disjoint specialization. Each subclass inherits the attributes of its superclass and adds its own.

Retail Chain EER — Design Walkthrough Explained

- **Core Entities**

STORE, EMPLOYEE, PRODUCT, CATEGORY, INVENTORY, CUSTOMER, TRANSACTION, and TRANS. LINE (a weak entity identified by Transaction + line number).

- **Key Relationships**

EMPLOYS (Store–Employee), STOCKS (Store–Product via Inventory), IN (Product–Category), MAKES (Customer–Transaction), CONTAINS (Transaction–Trans. Line).

- **Specialization: Employee**

EMPLOYEE is specialized (disjoint) into CASHIER, SALES ASSOCIATE, and STORE MANAGER — each inherits Name/ID/Store from Employee and adds its own attributes (e.g. Manager adds Authorization_Level).

- **Specialization: Product**

PRODUCT is specialized (disjoint) into PERISHABLE (adds Expiry_Date) and NON-PERISHABLE (adds Shelf_Life_Category), supporting different inventory rules per type.

- **Why EER Fits Here**

Plain ER cannot cleanly express that different employee/product types carry different attributes — generalization/specialization captures this naturally while avoiding repeated attribute definitions.